

Matteo Bombelli

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EDUCATION

Bachelor of Applied Science in Computing Science
Simon Fraser University

Sep. 2022 - Apr. 2027 (Expected)
Burnaby, BC

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, Java, Rust, HTML/CSS, SQL

Technologies: React, Next.js, FastAPI, Spring Boot, PostgreSQL, Linux

Cloud & Tools: AWS (S3), Azure (Entra ID/Graph API), Vercel, Docker, Git, Caddy

WORK EXPERIENCE

Technical Analyst Intern | LineZero, Vancouver, BC (Remote) May 2025 - Present

- Built and deployed a **retrieval-augmented chatbot** with FastAPI, Python, and PostgreSQL that answers questions from clients' internal knowledge bases, wiring in the Workvivo API for real-time chat and **text embeddings** for accurate search.
- Shipped **integrations** linking the company's employee-recognition platform to other SaaS apps, added **real-time notifications** and front-end features, and reviewed teammates' pull requests.
- Scoped and built bespoke client widgets and chatbots end to end, including a **context-aware widget** that surfaces employee updates and a **reusable RAG chat framework**, owning client discovery, security, and deployment for each one.
- Wrote internal Python tools to automate manual workflows, including a **Monday.com migration tool** that imported legacy Excel datasets and cut hours of repetitive data entry.
- Configured databases and managed cloud services across **Azure** (Entra ID and client identity), **AWS** (hosting for the RAG bot, recognition tool, and custom widgets), and **Vercel** (deployments).
- Onboarded and **trained new interns**, getting them productive on the team's stack and workflows.

Technician / Technology Associate | Staples Canada, Vancouver, BC Jul. 2021 - Mar. 2025

- Diagnosed and repaired **hardware and software** issues from intake to handoff, and hit **technology sales targets** by matching customers with the right IT solutions.

PROJECTS

WebTunes | Next.js, TypeScript, PostgreSQL, AWS S3 | [Live](#) | [Code](#) Jun. 2026

- Streamed each user's library straight from S3 to the browser via **presigned URLs** and **HTTP range requests**, so the Next.js server never proxies audio bytes.
- PostgreSQL **full-text search** over title, artist, album, and lyrics, backed by a generated tsvector and **GIN index**, with an LRCLIB lyrics fallback.
- Friend-based library sharing and playlists modeled in Drizzle ORM behind a single access check, with **Auth.js** sessions, shipped as an installable **PWA**.

URL Shortener & Analytics API | Spring Boot, Java, PostgreSQL, Redis, Nginx | [Code](#) Jan. 2026

- **Base62 encoding** generates short, collision-resistant links in a Spring Boot REST API, persisted through Spring Data JPA and PostgreSQL.
- **Cut redirect latency** by caching hot links in Redis behind an Nginx reverse proxy, with documented REST contracts.

Pipelined Reliable Transfer Protocol (P RTP) | Python, Wireshark | [Code](#) Nov. 2025

- Reliable transport over UDP in Python with a three-way handshake, **sliding-window pipelining**, and **congestion and flow control** via custom headers.
- Verified reliability with Wireshark traces and a **packet-loss fuzzer** against a non-blocking socket server.

Blackjack vs Agent | R, React, TypeScript | [Live](#) | [Code](#) Feb. 2025

- Trained a Monte Carlo **reinforcement learning** agent in R across 250,000 hands, raising win rate to **43.3%** and beating random play by 15.3 points.
- Built an interactive React site with sound and animation to play against the trained agent.